Water Supply & Fleet Billing Management System

# Requirements Specification

Functional and non-functional requirements

| | |
|---|---|
| **Prepared for** | Shree Karunambika Traders |
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# 1. User roles

| Persona | Works at | Main job | Key need | Biggest risk |
|---|---|---|---|---|
| **Admin** (owner or manager) | Office, phone | Controls users, rates, day reopening; reviews audit | See problems at a glance, trust the numbers | Undetected leakage of water or cash |
| **Dispatch Operator** | Filling point, busy and interrupted | Runs fills on three meters | Fast, large targets, few keystrokes | Wrong vehicle or reading during peak time |
| **Billing Clerk** | Office | Invoices, consolidated billing, statements | Find unbilled trips, print quickly | Duplicate or missed billing |
| **Accountant** | Office, periodic | Vouchers, ledgers, reconciliation, day close | Standard ledger formats, clean audit trail | Balances that do not tie out |

## Permission matrix

● full, ◐ limited as noted, – none.

| Capability | Admin | Operator | Billing Clerk | Accountant |
|---|---|---|---|---|
| Dashboard | ● | ◐ meters, trips | ◐ sales | ● |
| Supplier, Vehicle, Customer master | ● | ◐ view | ◐ view, add customer | ◐ view |
| Rates | ● | – | ◐ view | ◐ view |
| Meters setup and chain reset | ● | – | – | – |
| Live dispensing | ● | ● | ◐ view | ◐ view |
| Manual override, no amount change | ● | ◐ if granted | ◐ if granted | ● |
| Manual override, amount changes | ● | – | – | ● |
| Sales invoice create | ● | ● | ● | ◐ view |
| Invoice cancel | ● | – | ◐ same day, unpaid | ● |
| Consolidated invoice, fleet statement | ● | – | ● | ● |
| Receipt voucher | ● | ◐ cash only | ● | ● |
| Payment voucher | ● | – | – | ● |
| Ledger Master | ● | – | – | ● |
| Ledgers and statements | ● | – | ◐ party statements | ● |
| Reports | ● | ◐ own day | ◐ sales | ● |
| Day Opening | ● | ● | – | ● |
| Day Closing | ● | – | – | ● |
| Reopen closed day | ● | – | – | – |

| Capability | Admin | Operator | Billing Clerk | Accountant |
|---|---|---|---|---|
| Users, roles, audit log | ● | – | – | – |

# 2. Functional requirements

## Master data

| ID | Requirement |
|---|---|
| FR-VM-01 | Create, view, edit and deactivate suppliers: Supplier Name*, Contact Person, Phone No.*, Alternate Phone No., Address, Status. |
| FR-VM-02 | Create, view, edit and deactivate vehicles: Vehicle No.*, Supplier*, Capacity (Litres)*, Trip Cost*, Status*, plus Nickname, Rate, Billing Basis, Unit of Measurement, Water Details, Driver Name, Driver Phone. |
| FR-VM-03 | Every vehicle gets a unique system code, one letter plus a number (e.g. `A13`), proposed by the system. Admin may change it only before first use. Codes are never reused. |
| FR-VM-04 | Vehicle No. is stored uppercase with spaces tidied, and must be unique. The format check allows short forms such as `TN 67 C`, because the designs use them. |
| FR-VM-05 | Nickname defaults to supplier short name + plate, e.g. `Sangeetha TN 38 M 3`. |
| FR-VM-06 | One supplier has many vehicles. A vehicle has one supplier at a time. Transfer is an Admin action, keeps history, and needs a zero or carried balance decision. |
| FR-VM-07 | Rates are dated. A new rate starts from a chosen date, and past invoices keep the rate they were saved with. |
| FR-VM-08 | Delete is only allowed for a record with no history. Otherwise the action offers Inactive. |
| FR-VM-09 | Inactive vehicles cannot start a fill. Inactive suppliers make all their vehicles unavailable, without changing the vehicles' own status. |
| FR-VM-10 | Search by Vehicle No., and filter by Active, Inactive or All, as designed. Search also matches code, nickname and supplier. |
| FR-VM-11 | Customers and staff are maintained as separate party types, each with their own ledger. |
| FR-VM-12 | Saving a supplier, customer or staff member with an opening balance posts one opening entry, once. |

## Meter chain

| ID | Requirement |
|---|---|
| FR-MTR-01 | The system has exactly three meters: Meter 1, Meter 2, Meter 3. Each has unit, decimals, maximum reading and status. |
| FR-MTR-02 | Each meter holds its own chain. Meters operate independently and concurrently. |
| FR-MTR-03 | A meter can have at most one open leg. A second attempt fails and shows which vehicle is filling. |
| FR-MTR-04 | When a leg starts, start reading = previous leg's end reading, and chain start time = previous leg's chain end time. Both are locked and cannot be typed. |
| FR-MTR-05 | Volume = end reading − start reading, in the meter's unit, to one decimal. |
| FR-MTR-06 | Starting a fill is the operator's confirmation that the dial shows the locked reading. A "Dial differs" action records the actual dial value. |

| ID | Requirement |
|---|---|
| FR-MTR-07 | If the dial is above the locked reading, an Unattributed leg for the difference is added to the chain and the fill continues. |
| FR-MTR-08 | If the dial is below the locked reading, the meter moves to Fault. No fill can start until an Admin resolves it. |
| FR-MTR-09 | An Unattributed leg must be assigned to a vehicle, or written off by an Admin with a reason, before day close. |
| FR-MTR-10 | If final reading is below start reading, the system asks whether the meter rolled over. Rollover is allowed only when start is within 1% of maximum. Volume = (maximum + one display step − start) + end. |
| FR-MTR-11 | Replacing or recalibrating a meter is an Admin **chain reset**: it records final old reading, new starting reading and reason, and starts a new chain version. |
| FR-MTR-12 | Meter status is one of Idle, Filling, Fault, Offline. Status changes reach all open screens within 2 seconds. |

## Dispensing

| ID | Requirement |
|---|---|
| FR-DSP-01 | A session records vehicle, supplier, In Time, Out Time and 1 to 3 meter legs. |
| FR-DSP-02 | In Time and Out Time are stamped from the server clock when fills start and stop. Legs on one session may start and stop at different times: In Time is the earliest start, Out Time the latest stop. |
| FR-DSP-03 | A vehicle can have only one open session. |
| FR-DSP-04 | The Meter Readings block always lists all three meters. An unused meter shows 0.0 and creates no leg. |
| FR-DSP-05 | Session total volume = sum of its legs. |
| FR-DSP-06 | If total volume exceeds tank capacity by more than the tolerance (default 5%), the system warns and needs a reason before completing. |
| FR-DSP-07 | A leg stopped with zero flow can be aborted. The aborted leg stays in the log but does not advance the chain. |
| FR-DSP-08 | Business date is set by In Time and the cut-off. Before 18:00 it is the same day; from 18:00 it is the next day. A session that crosses midnight or 18:00 is never split. |
| FR-DSP-09 | Each leg records how it was captured: Device, Operator Live, or Manual Override. |

## Manual override

| ID | Requirement |
|---|---|
| FR-OVR-01 | Manual entry or edit of a dispensing record is possible only from 18:00:00 to 23:59:59 server time, IST. Both bounds are settings. |
| FR-OVR-02 | Outside the window, In Time, Out Time, start reading and saved end readings are read-only for every role, including Admin. |
| FR-OVR-03 | The final reading of an **open** leg is live capture, not an override, and is allowed all day. |
| FR-OVR-04 | Requires permission, a reason code (Operator Error, Power or Network Outage, Meter Display Fault, Missed Entry, Party Dispute, Other) and a note of at least 15 characters. |
| FR-OVR-05 | The window is checked again on the server at the moment of saving. If it has closed, the save fails and the entered values stay on screen. |

| ID | Requirement |
|---|---|
| FR-OVR-06 | Changing a leg's end value moves the next leg's locked start to match. New end must not exceed the next leg's end. Only one following leg is affected, because its own end stays the same. |
| FR-OVR-07 | Before saving, an impact preview lists every record and amount that will change. |
| FR-OVR-08 | If the change alters a posted amount, the voucher is reversed and a corrected one posted, and the user must hold the amount-change permission. Trip-billed invoices normally have no amount change. |
| FR-OVR-09 | Records in a closed day cannot be overridden until an Admin reopens the day. |
| FR-OVR-10 | Every override is written to the audit log with before values, after values, reason, user, time and device. |
| FR-OVR-11 | An overridden record shows a MANUAL badge permanently, on screen and on the override report. |
| FR-OVR-12 | A network outage is handled by paper record, then entered in the window with reason Power or Network Outage. |

## Invoicing

| ID | Requirement |
|---|---|
| FR-INV-01 | Invoice number is assigned at save, from one gapless series, starting from the agreed number. The screen shows the expected next number until then. |
| FR-INV-02 | Selecting Vehicle Code or Vehicle Name fills the other, and Vehicle Supplier and Rate. |
| FR-INV-03 | Trip defaults to 1. Amount = Trip × Rate for TRIP vehicles, or Volume × Rate for VOLUME vehicles, rounded half-up to 2 decimals. |
| FR-INV-04 | Rate defaults from the vehicle's dated rate. Changing it requires permission and is logged. |
| FR-INV-05 | Payment Mode is Cash or Credit. |
| FR-INV-06 | Item Details shows S.No, Material, Qty, Measurement, Rate, Total Amount. |
| FR-INV-07 | Save creates the invoice and its ledger postings together. If either fails, neither is kept. |
| FR-INV-08 | Bill To defaults to the vehicle's supplier and can be set to a customer. The ledger posts to Bill To, tagged with the vehicle. |
| FR-INV-09 | A session can be on only one active invoice. |
| FR-INV-10 | Consolidated invoice: choose supplier and period, select unbilled sessions, and produce one invoice with one line per session. |
| FR-INV-11 | Fleet statement: choose supplier and period, and list invoices already raised, with no ledger effect. |
| FR-INV-12 | Cancel requires reason and reverses the postings. The number is kept and the invoice is shown as Cancelled. Allocations must be released first. |
| FR-INV-13 | Editing a posted invoice is Cancel + new invoice. Editing is not possible in place. |
| FR-INV-14 | Previous and Next move through invoices, View opens read-only, and Print outputs the PDF. Editing dispensing fields follows FR-OVR. |
| FR-INV-15 | Invoice Display lists invoices, receipts and payments for a period with Cash, Credit, Receipt and Payment columns, plus Summary and Opening/Closing balance. |
| FR-INV-16 | Closing Balance on Invoice Display = Opening + Cash Sales + cash Receipts − cash Payments. |

## Transactions

| ID | Requirement |
|---|---|
| FR-TXN-01 | Receipt Voucher: Receipt No., Date, Party / Supplier*, Mode of Receipt, Amount*, Reference No, Narration. |
| FR-TXN-02 | Payment Voucher: Voucher No., Payment Date, Ledger*, Pay To (when applicable), Vehicle (optional), Payment Mode, Amount*, Reference No, Narration. |
| FR-TXN-03 | Modes are Cash, Bank, UPI. Reference No is required for Bank and UPI. |
| FR-TXN-04 | Receipts and payments have separate gapless series. In ledgers they display as `RV-432` and `PV-432`, because the designs show both as 432. |
| FR-TXN-05 | Amount must be greater than zero. |
| FR-TXN-06 | Saving posts immediately per the posting rules. |
| FR-TXN-07 | The pane shows the party's or ledger's balance now and after this voucher, updating as the user types. |
| FR-TXN-08 | A cash payment above the available cash balance needs a warning and confirmation. |
| FR-TXN-09 | A receipt can be allocated to open invoices, oldest first by default, adjustable by hand. Any remainder stays as Unallocated. |
| FR-TXN-10 | A vehicle-level allocation updates that vehicle's ledger. |
| FR-TXN-11 | Editing or deleting a voucher is Cancel + new voucher, with a reversal. |
| FR-TXN-12 | Receipt History and Payment History can be filtered by date, party, mode and status. |
| FR-TXN-13 | Ledger Master creates ledgers (e.g. Vehicle Maintenance, Fuel Expense, Oil Change, Others). System ledgers cannot be changed or removed. Ledgers with entries cannot be deleted. |

## Ledger and posting

| ID | Requirement |
|---|---|
| FR-LED-01 | Every financial event posts balanced lines: total debit equals total credit. |
| FR-LED-02 | Ledger lines are append-only. They are never updated or deleted by the application. |
| FR-LED-03 | Each line stores Date (when posted), Value Date (the business date it belongs to), Particulars, Voucher/Ref No, Debit, Credit and Running Balance. |
| FR-LED-04 | Running balance is stored at posting and follows posting order, so a late correction never rewrites past lines. It appears with a Dr or Cr suffix. |
| FR-LED-05 | Statements include Opening Balance b/f, entries, period Debit and Credit sub-totals, Closing Balance c/f and a reconciliation line. |
| FR-LED-06 | Statements default to Date order. A "Sort by Value Date" option recalculates the displayed balance without changing stored data. |
| FR-LED-07 | Vehicle ledger shows lines tagged to the vehicle. It is a view, not a separate account, so nothing posts twice. |
| FR-LED-08 | Sum of vehicle balances + unallocated = supplier balance, always. |
| FR-LED-09 | Reversal lines swap debit and credit, cite the original voucher, and post on the date of correction with the original value date. |
| FR-LED-10 | Posting to a closed business date is not possible. Corrections post to the open day and quote the original value date. |

| ID | Requirement |
|---|---|
| FR-LED-11 | Statements export to PDF and Excel with the same columns. |

### Posting rules

"Party" is the receivable or staff account, tagged with the supplier, customer or staff member, and the vehicle where known.

| Code | Event | Debit | Credit |
|---|---|---|---|
| P1 | Credit sale | Party | Water Sales |
| P2 | Cash sale | Party, then Cash in Hand | Water Sales, then Party |
| P3 | Receipt | Cash, Bank or UPI | Party |
| P4 | Operational expense payment | Expense ledger | Cash, Bank or UPI |
| P5 | Payout or refund to party | Party | Cash, Bank or UPI |
| P6 | Staff advance | Staff Advances (staff) | Cash, Bank or UPI |
| P7 | Staff advance recovered | Cash, Bank or UPI | Staff Advances (staff) |
| P8 | Cash short at closing | Cash Over/Short | Cash in Hand |
| P9 | Cash over at closing | Cash in Hand | Cash Over/Short |
| P10 | Opening balance | Party or account | Opening Balance Equity (reversed if a credit balance) |
| P11 | Any cancellation | Original credits | Original debits |

P2 posts through the party on purpose. Cash trips then appear on the supplier's statement, netting to zero, and "Received Amount" per vehicle is correct without any special handling.

## Day operations

| ID | Requirement |
|---|---|
| FR-DAY-01 | Day Opening shows opening cash carried from the last close. |
| FR-DAY-02 | View Day Sales shows Day Opening, Daily Transactions and Day Closing, with Opening Balance, Closing Balance, Actual Cash Balance and Cash Difference. |
| FR-DAY-03 | Day Closing is blocked while any of these exist for the day: an open leg, an unattributed leg, or a completed session without invoice or Bill Later status. |
| FR-DAY-04 | Cash Difference = Actual Cash Balance − Closing Balance. A non-zero difference needs a reason and posts P8 or P9. |
| FR-DAY-05 | Closing locks the business date and carries Actual Cash Balance to the next opening. |
| FR-DAY-06 | Only Admin can reopen a day, with a reason. Reopening a day that has later closed days requires reopening them in reverse order. |

## Dashboard

| ID | Requirement |
|---|---|
| FR-DASH-01 | Cards as designed: Total Sales (This Month), Today's Sales, Today's Receipt, Today's Payment. |
| FR-DASH-02 | Second row: Cash Sales, Credit Sales, Total Water Trips, Outstanding Amount, Opening and Closing Cash Balance. |
| FR-DASH-03 | Live tiles for Meters 1, 2 and 3: status, vehicle, reading, elapsed time. Selecting one opens the desk with that meter selected. |
| FR-DASH-04 | Monthly Sales chart covers the last 6 months. Selecting a bar opens Invoice Display for that month. |
| FR-DASH-05 | Quick Access: Sales Invoice, Invoice Display, Vehicle Master, Reports. Each opens its screen directly. |
| FR-DASH-06 | Recent Invoices shows the last 3, with View All opening Invoice Display. |
| FR-DASH-07 | An alert strip shows unattributed legs, blocked meters, unbilled sessions and whether the override window is open. |
| FR-DASH-08 | The month selector in the header changes month-scoped figures. Today's figures stay on today. |

## Reports

| ID | Requirement |
|---|---|
| FR-RPT-01 | Reports from the requirements PDF: Daily Sales, Date-wise Sales, Supplier-wise Sales. |
| FR-RPT-02 | Operational report: Vehicle Name, Registration Number, Total Volume, Unit Rate, Total Bill Amount, Received Amount, Outstanding Net Balance, per vehicle with a supplier subtotal and grand total. |
| FR-RPT-03 | Ageing in buckets 0–7, 8–15, 16–30, 31–60, 60+ days, by invoice value date, using allocations. |
| FR-RPT-04 | Accounts: Supplier-wise Outstanding, Staff Outstanding, Customer Outstanding, Trial Balance, Cash Book. |
| FR-RPT-05 | Control: meter reconciliation, unattributed legs, override report, cancelled documents, rate changes. |
| FR-RPT-06 | All reports filter by date range, supplier and vehicle, and export to PDF and Excel. |
| FR-RPT-07 | Report totals must equal ledger totals for the same filter. |

## Admin and audit

| ID | Requirement |
|---|---|
| FR-ADM-01 | Admin creates users, assigns a role and grants optional permissions. Users can be deactivated, never deleted. |
| FR-ADM-02 | Staff must change a temporary password at first login. |
| FR-ADM-03 | Logged events: login success, failure and logout; user and permission changes; master changes; rate changes; dispensing starts and stops; variances; overrides; invoices; vouchers; cancellations; day open, close and reopen; exports; chain resets. |
| FR-ADM-04 | Each entry records time, user, role, action, record, before and after values, reason, IP and device. |
| FR-ADM-05 | The log has no edit or delete function for any role. |
| FR-ADM-06 | Each entry includes a hash of the previous entry, so any change breaks the chain. Admin can run a verification. |

| ID | Requirement |
|---|---|
| FR-ADM-07 | Only Admin can view the log. Viewing and exporting it is itself logged. |

# 3. Non-functional requirements

| Area | ID | Requirement |
|---|---|---|
| **Security** | NFR-SEC-01 | HTTPS only, TLS 1.2 or newer. Data encrypted at rest. |
| | NFR-SEC-02 | Passwords stored with a modern slow hash. Lockout after 5 failed attempts for 15 minutes. |
| | NFR-SEC-03 | Two-step sign-in required for Admin and Accountant. |
| | NFR-SEC-04 | Sessions time out after 30 minutes idle. Dispatch desks may use 8 hours, as they are shared stations. |
| | NFR-SEC-05 | Permissions are enforced on the server, on every request. Hiding a button is not treated as security. |
| | NFR-SEC-06 | The application's database account cannot update or delete ledger or audit rows. |
| **Concurrency** | NFR-CON-01 | Three meters run at once, with independent operators, with no blocking between meters. |
| | NFR-CON-02 | Two simultaneous starts on one meter: exactly one succeeds. |
| | NFR-CON-03 | Simultaneous saves never produce duplicate or skipped numbers. |
| | NFR-CON-04 | Simultaneous postings for one party give a correct running balance. Balance rows are locked in a fixed order to prevent deadlocks. |
| | NFR-CON-05 | Two users editing one record: the second save is refused with a "changed by" message. |
| | NFR-CON-06 | A repeated request (double click, network retry) has no second effect. |
| **Performance** | NFR-PER-01 | Start and stop fill: under 300 ms at the 95th percentile. |
| | NFR-PER-02 | Invoice save with posting: under 500 ms at the 95th percentile. |
| | NFR-PER-03 | Meter status reaches all screens within 2 seconds. |
| | NFR-PER-04 | A one-year statement opens within 2 seconds, and a full-year report within 5 seconds. |
| | NFR-PER-05 | Dashboard loads within 1.5 seconds. |
| **Reliability** | NFR-REL-01 | 99.5% availability in operating hours. |
| | NFR-REL-02 | Point-in-time recovery with at most 5 minutes of data lost, and service restored within 1 hour. |
| | NFR-REL-03 | Daily backups kept 35 days, monthly kept 7 years. Restore tested quarterly. |
| | NFR-REL-04 | During a site internet outage, staff follow the paper fallback (FR-OVR-12). A local server is the option if outages are frequent. |
| **Audit** | NFR-AUD-01 | Audit written in the same transaction as the action. |
| | NFR-AUD-02 | Audit kept at least 8 years. Daily copy to write-once storage. Please confirm the period with your auditor. |
| | NFR-AUD-03 | Server clock synchronised by NTP. All times stored in UTC, displayed in IST. |
| **Usability** | NFR-USE-01 | Desk screens work by keyboard alone. Targets at least 44 px on touch screens. |

| Area | ID | Requirement |
|---|---|---|
| | NFR-USE-02 | Indian formats: `4,94,784.00`, `DD/MM/YYYY`, 12-hour time. |
| | NFR-USE-03 | Meets WCAG 2.2 AA for contrast, focus and labels. |
| **Scale** | NFR-SCA-01 | Sized for 1,000 sessions a day and 10 years of ledger data without redesign. |

# 4. Recommended technology

| Layer | Recommendation | Why |
|---|---|---|
| **Frontend** | React with TypeScript | Large talent pool, strong for form and table heavy apps |
| **Server state** | TanStack Query | Caching and refetching. Server stays the source of truth for money. |
| **Local state** | Zustand | Light. Only for UI state such as selected meter |
| **Forms** | React Hook Form with Zod | Handles the dense invoice form well |
| **Data tables** | TanStack Table | Ledgers and reports |
| **Charts** | Apache ECharts | Monthly Sales and analytics |
| **Offline shell** | PWA | Installs on desk PCs, with fast reloads |
| **Backend** | NestJS (TypeScript), modular monolith | Shared language with frontend, clear module boundaries |
| **Live updates** | Server-Sent Events | One-way meter status is simple and enough. WebSocket is unnecessary. |
| **Database** | PostgreSQL 16 or newer | Exact decimal arithmetic, partial unique indexes, overlap constraints and row locks directly enforce chain rules |
| **Background jobs** | pg-boss, on PostgreSQL | PDF and audit export jobs, with no extra infrastructure |
| **Cache** | None at first | Scale does not need it. Add Redis only if measurements show a need. |
| **PDF** | Server-side HTML to PDF | Print matches screen |
| **Hosting** | AWS Mumbai (`ap-south-1`): containers on ECS Fargate, RDS PostgreSQL Multi-AZ, S3 with Object Lock | Low latency to Coimbatore, Indian data location, write-once audit archive |
| **Monitoring** | CloudWatch with Sentry | Errors, performance and alerts |
| **Delivery** | GitHub Actions with staging and production | Repeatable, with migrations reviewed |